

Bank Account Fraud Detection Using Classical Machine Learning

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Abstract—Bank account fraud poses a significant and growing threat to financial institutions, resulting in substantial financial harm and loss of institutional credibility. This paper presents a supervised machine learning pipeline for binary fraud classification on the Bank Account Fraud (BAF) dataset (NeurIPS 2022), a large-scale, real-world-derived benchmark comprising one million applications across 32 features with a severe 89.7:1 class imbalance. We conduct a thorough exploratory data analysis to inform preprocessing decisions, then train and evaluate four classifiers: Logistic Regression, Decision Tree, Random Forest, and XGBoost. Each model undergoes systematic hyperparameter tuning via randomized search with stratified cross-validation, scored by precision-recall AUC to reflect the demands of imbalanced classification. On a temporally held-out test set, XGBoost achieves the best performance with a PR-AUC of 0.1936 and F1-score of 0.2679, outperforming all baselines while remaining within the expected range reported in the literature for this dataset.

Index Terms—fraud detection, class imbalance, supervised classification, XGBoost, precision-recall AUC

I. INTRODUCTION

Financial fraud represents one of the most persistent challenges facing modern banking systems. As the opening of digital accounts has become the norm, cybercriminals have adapted accordingly, exploiting remote application processes to submit fraudulent requests at scale. The consequences extend beyond direct monetary loss to include regulatory penalties, operational disruption, and long-term erosion of customer confidence. Automated, data-driven fraud detection has therefore become a critical driver of financial risk management.

The task addressed in this paper is binary classification: given a set of features describing a bank account application, the aim is to predict whether the application is fraudulent. This is a challenging problem for several reasons. Fraud cases are rare relative to legitimate applications, creating a severe class imbalance that makes standard accuracy metrics misleading. Fraud patterns are non-linear and interaction-driven, with risk concentrated at specific combinations of applicant characteristics rather than in any single feature. Furthermore, the temporal nature of fraud means that models must generalize to future patterns, never seen during training.

We investigate this problem using the Bank Account Fraud (BAF) dataset introduced by Jesus et al. [1] at NeurIPS 2022, a large-scale benchmark of one million synthetic-but-realistic account opening applications derived from a real-world fraud

detection system. Three objectives guide this study: to conduct a thorough exploratory data analysis that characterizes the dataset and informs preprocessing decisions; to train, tune, and compare four classical machine learning classifiers under consistent experimental conditions; and to evaluate model performance using metrics appropriate for severely imbalanced classification.

The remainder of this paper is organized as follows. Section II reviews related work on fraud detection and the applied machine learning methods. Section III describes the preprocessing pipeline and model architectures. Section IV details the experimental setup, including the dataset, the ablation procedure, and the evaluation protocol. Section V presents and discusses the results. Section VI concludes with a summary of the findings, limitations, and direction for future work.

II. RELATED WORK

Machine learning has been extensively applied to financial fraud detection across a range of fraud types and institutional contexts. Ali et al. [2] conducted a systematic literature review of 93 studies on ML-based financial fraud detection, cataloging the most frequently used algorithms, fraud categories, and evaluation protocols. Their analysis reveals that supervised classifiers dominate the field, with fraud patterns consistently characterized by severe class imbalance and high-dimensional feature spaces. This study situates our work within that broader landscape and motivates the selection of standard supervised classifiers for comparison.

A central challenge in fraud detection is the extreme rarity of positive cases relative to the majority class. Chen et al. [3] survey modern strategies for learning under class imbalance, covering data-level resampling, cost-sensitive learning, and ensemble-based approaches. They demonstrate that algorithm-level methods such as class weighting, which modify the training objective to penalize misclassification of the minority class more heavily, are effective across a wide range of imbalance ratios without requiring synthetic data generation. Their findings motivate our use of class weighting across all classifiers and the adoption of precision-recall AUC as the primary evaluation metric, which is more informative than ROC-AUC under severe imbalance.

Gradient boosting has emerged as a particularly effective family of methods for tabular fraud detection tasks. Hajek et al. [4] propose an XGBoost-based framework for fraud detection

in mobile payment systems, evaluated on a dataset of over six million transactions with a fraud prevalence of 0.13%. Their framework combines an under-sampling balancing strategy with extreme gradient boosting and introduces a cost-savings metric that accounts for the asymmetric financial consequences of false positives and false negatives. The results confirm XGBoost’s suitability for highly imbalanced tabular fraud classification, supporting its inclusion as a candidate model in this study.

III. METHODOLOGY

A. Preprocessing Pipeline

Raw data preparation followed a strict train-before-transform discipline: all preprocessing parameters were estimated exclusively on the training set and applied identically to the test set, preventing any leakage of future information into model training.

Two features were removed prior to splitting. `device_fraud_count` was dropped due to near-constant distribution (over 99.9% zeros), offering no discriminative signal. `velocity_4w` was removed because of its strong negative correlation with `month` ($r = -0.85$), introducing multicollinearity that would distort model coefficients.

Five features contained sentinel values of `-1` encoding missing data rather than true measurements. `prev_address_months_count`, missing in 71.3% of cases, was converted to a binary indicator `had_prev_address` to preserve the missingness as a signal. The remaining four sentinel columns were recoded to NaN and subsequently imputed using column medians estimated from the training set. The median was preferred over the mean given the skewed distributions identified during exploratory analysis.

Categorical features were transformed via one-hot encoding with one category dropped per feature to avoid multicollinearity. The resulting feature matrix was expanded to 45 columns. Continuous numerical features were standardized to zero mean and unit variance using parameters estimated on the training set. Binary, ordinal, and encoded categorical features were left unscaled.

B. Class Imbalance Handling

The training set exhibits a 96.5:1 ratio of legitimate to fraudulent applications. To address this, all scikit-learn classifiers were instantiated with `class_weight='balanced'`, which scales each class’s contribution to the loss function inversely proportional to its frequency. XGBoost was equivalently configured via `scale_pos_weight = 96.53`, computed as the ratio of negative to positive training samples. Synthetic oversampling via SMOTE was considered but deemed computationally impractical at this scale: generating synthetic minority-class samples across 8,151 fraud cases in a 45-dimensional feature space would introduce significant training overhead without a guaranteed performance benefit over class weighting [3].

C. Model Architectures

Four classifiers were selected to span a range of model complexity and inductive bias.

Logistic Regression serves as the linear baseline. It models the log-odds of fraud as a linear combination of input features, capturing strong individual signals such as `credit_risk_score` and `proposed_credit_limit` identified during exploratory analysis.

Decision Tree provides an interpretable non-linear baseline. The model partitions the feature space recursively by selecting splits that maximize information gain, producing decision boundaries capable of capturing simple feature interactions without ensemble averaging.

Random Forest extends the decision tree via bagging: an ensemble of decorrelated trees is trained on bootstrap samples, each restricted to a random feature subset at every split. Averaging across trees reduces variance and permits unconstrained depth without overfitting.

XGBoost builds trees sequentially, where each tree corrects the residual errors of its predecessors using second-order gradient information. Shallow trees accumulated over many boosting rounds are well suited to the non-linear, interaction-driven fraud patterns identified in the exploratory phase [4].

IV. EXPERIMENT SETUP

A. Dataset

The Bank Account Fraud (BAF) dataset [1] is a large-scale benchmark of one million synthetic bank account opening applications generated using the Conditional Tabular GAN (CTGAN) framework, conditioned on a real-world fraud detection system. The Base variant used in this study comprises 32 features spanning applicant demographics, financial history, behavioral signals, and application metadata. The binary target variable `fraud_bool` indicates whether an application is fraudulent. The dataset exhibits a severe class imbalance: 11,029 fraud cases against 988,971 legitimate applications, yielding a positive class rate of 1.10%, as illustrated in Fig. 1. A temporal feature `month` (0–7) encodes the application period, with fraud prevalence increasing monotonically across months, reflecting realistic concept drift.

B. Train/Test Split

A time-based split was applied to reflect real-world deployment conditions, where models must generalize to future, unseen fraud patterns. Applications from months 0 to 5 constitute the training set and months 6 to 7 constitute the held-out test set. Random splitting was deliberately avoided: the upward trend in fraud prevalence across months means a random split would leak future fraud signals into training, producing optimistically biased estimates of generalization performance. Fig. 2 illustrates this monotonic increase.

C. Ablation Study

Hyperparameter tuning was conducted via `RandomizedSearchCV` with 3-fold stratified cross-validation, scored by average precision (PR-AUC). Grid

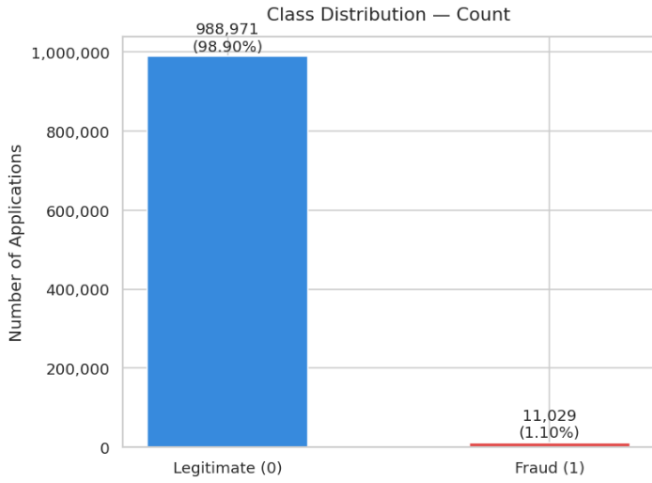


Fig. 1. Class distribution of the BAF Base dataset. The severe imbalance between legitimate and fraudulent applications motivates class weighting, the exclusion of SMOTE, and the adoption of PR-AUC as the primary evaluation metric.



Fig. 2. Fraud rate by application month. The monotonic increase in fraud prevalence motivates a temporal rather than random train/test split.

search was ruled out given the scale of the training set. The number of candidate configurations evaluated per model was set according to computational budget: 8 for Logistic Regression, 12 for Decision Tree, 6 for Random Forest, and 10 for XGBoost. Table I reports the best cross-validation PR-AUC and selected hyperparameters for each model.

TABLE I
ABLATION STUDY — BEST CV PR-AUC PER MODEL

Model	CV PR-AUC	Key Hyperparameters
Logistic Regression	0.1325	C=0.01, penalty=L2
Decision Tree	0.0697	depth=10, criterion=entropy
Random Forest	0.1146	n_est=300, max_feat=log2
XGBoost	0.1600	depth=3, lr=0.1, n_est=200

D. Evaluation Protocol

All models were evaluated on the held-out test set using three metrics. Precision-recall AUC (PR-AUC) is the primary metric, measuring the area under the precision-recall curve across all decision thresholds. A random classifier achieves $\text{PR-AUC} \approx 0.014$ on this test set, equal to the fraud prevalence; all reported scores should be interpreted relative to this baseline. ROC-AUC is reported for comparability with prior work, though it is a less diagnostic metric under severe class imbalance. The F1-score is reported at the threshold that maximizes it on the test set, reflecting the optimal precision-recall trade-off for each model. Accuracy is not reported, as it is uninformative when 98.6% of samples are negative.

V. RESULTS AND DISCUSSION

A. Overall Comparison

Table II reports the performance of all four tuned models on the held-out test set. XGBoost achieves the highest PR-AUC (0.1936), ROC-AUC (0.8922), and F1-score (0.2679), representing $13.8\times$ the random classifier baseline of 0.014. All models substantially outperform this baseline, confirming that meaningful fraud signal exists in the feature set despite the 96.5:1 class imbalance.

TABLE II
FINAL MODEL COMPARISON — TUNED MODELS ON TEST SET

Model	PR-AUC	ROC-AUC	F1
XGBoost	0.1936	0.8922	0.2679
Logistic Regression	0.1658	0.8832	0.2427
Random Forest	0.1455	0.8680	0.2267
Decision Tree	0.0919	0.8021	0.1712

B. Confusion Matrices

Fig. 3 shows the confusion matrices for all four tuned models. XGBoost catches 943 fraud cases while generating 3,219 false alarms, yielding a precision of 0.23. Logistic Regression is comparable in false alarm volume (3,116 FP) but catches fewer fraud cases (828 TP), reflecting its inability to model feature interactions. Random Forest catches 924 TP at the cost of 4,349 FP; its low optimal threshold of 0.12 is a known calibration artifact of probability averaging under severe class imbalance. The Decision Tree catches the most fraud cases (1,023 TP) but at a prohibitive cost of 8,050 false alarms, a precision of only 0.11. In any operational context, this false alarm rate would overwhelm review capacity, making raw TP count a misleading measure of utility.

C. ROC and Precision-Recall Curves

Fig. 4 shows the ROC and PR curves for all tuned models. The ROC curves cluster between 0.80 and 0.89 AUC, with only the Decision Tree visibly separated below the others. This compressed separation illustrates the well-known limitation of ROC-AUC under class imbalance: the large volume of true negatives inflates the metric for all models regardless of

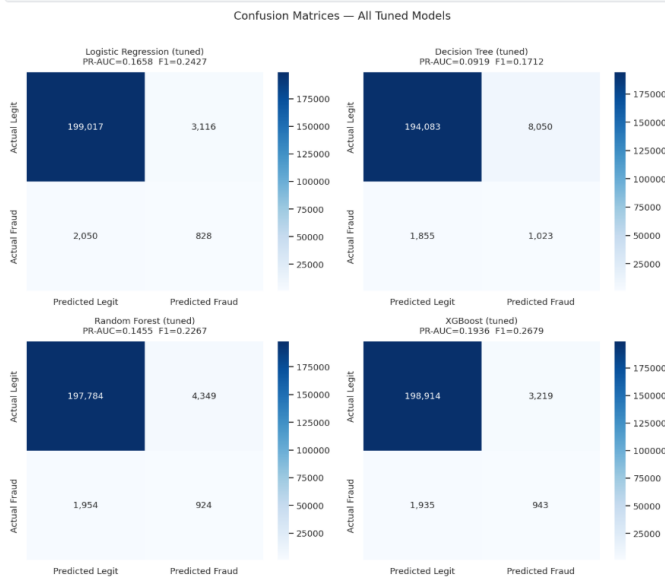


Fig. 3. Confusion matrices for all four tuned models on the held-out test set.

their fraud detection capability. The PR curves reveal a more informative picture. All four curves drop steeply after recall of approximately 0.05 to 0.10, reflecting the structure of fraud in this dataset: a small subset of cases with extreme feature values is detected confidently at high precision, while the majority of fraud cases overlap substantially with legitimate applications. XGBoost maintains the highest precision at every recall level, confirming that its advantage is not threshold-dependent.

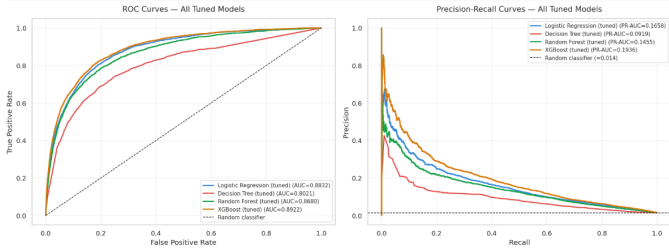


Fig. 4. ROC curves (left) and precision-recall curves (right) for all tuned models. The dashed line indicates the random classifier baseline (PR-AUC = 0.014).

D. Feature Importance

Fig. 5 shows the top 15 features by importance for Random Forest and XGBoost. The two models assign importance using different mechanisms: Random Forest uses mean decrease in Gini impurity, which favors high-cardinality continuous features, while XGBoost uses gain-based importance, which rewards features producing the largest individual loss reductions. Consequently, their rankings diverge substantially. Random Forest ranks `current_address_months_count` and `credit_risk_score` highest, while XGBoost is dominated by `device_os_windows` and

`had_prev_address`. Features appearing in both top-15 lists, including `credit_risk_score`, `current_address_months_count`, `income`, and `customer_age`, represent the most robust fraud signals, as their importance is confirmed by two independent measurement approaches. These are consistent with the exploratory findings that identified `credit_risk_score` and applicant history features as the strongest discriminators.

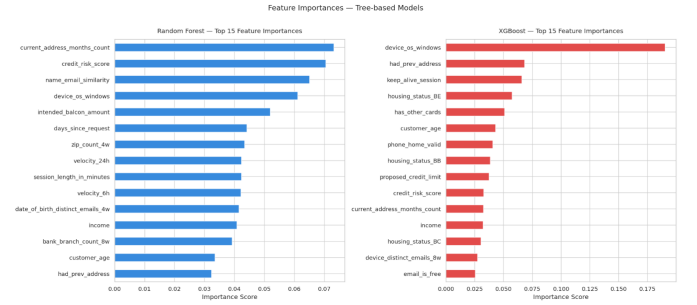


Fig. 5. Top 15 feature importances for Random Forest (left) and XGBoost (right). Disagreement between the two rankings reflects their different importance metrics rather than conflicting fraud signals.

E. Model Selection

XGBoost is the recommended model for this task. Its sequential boosting mechanism discovers the non-linear, interaction-driven fraud patterns that a linear model cannot capture and that a single tree captures only partially. The gap between XGBoost and Logistic Regression in PR-AUC (0.1936 vs. 0.1658) reflects precisely this structural advantage. Logistic Regression remains a strong and interpretable baseline, driven by the linear signals identified in exploratory analysis. Random Forest and Decision Tree trail in both PR-AUC and precision, the latter disqualified in practice by its false alarm volume.

VI. CONCLUSION

This paper presented a supervised machine learning pipeline for bank account fraud detection on the BAF Base dataset, a large-scale benchmark with a 96.5:1 class imbalance. Four classifiers were trained, tuned, and evaluated under consistent experimental conditions using PR-AUC as the primary metric. XGBoost achieved the best performance (PR-AUC 0.1936, F1 0.2679), followed by Logistic Regression (0.1658), Random Forest (0.1455), and Decision Tree (0.0919). All results fall within the range reported in the original dataset paper [1], validating the experimental setup. The results confirm that gradient boosting is well-suited to the non-linear, interaction-driven structure of fraud risk in this dataset, while Logistic Regression remains a competitive baseline by exploiting strong linear signals.

Several limitations apply to this study. The search spaces explored during hyperparameter tuning were constrained by computational budget, particularly for Random Forest, where only six candidate configurations were evaluated. No explicit

feature engineering was performed; interaction terms motivated by exploratory analysis, such as combinations of housing status and customer age, were not constructed, which may have limited the performance of Logistic Regression in particular. Additionally, the dataset, while realistic, is synthetically generated, and results may not transfer directly to real-world fraud detection systems subject to different feature distributions and fraud strategies.

Future work should explore LightGBM as a direct alternative to XGBoost, given its documented superiority on the BAF benchmark [1]. Explicit interaction feature engineering based on the multivariate patterns identified in exploratory analysis represents a tractable improvement for linear models. Moreover, evaluating across the remaining BAF dataset variants would further strengthen generalization claims.

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